NorthPay

Canadian payroll math reference · Tax year 2026

This document explains exactly how every paystub and T4 is calculated. Every formula, every bracket, every cap is here. If something in the app doesn't match this document, it's a bug.

Key 2026 changes (vs 2025)

What	2025	2026	Impact
Federal lowest bracket rate	15.00 %	14.00 %	Lower tax on every paystub
Federal BPA (high)	$16,129	$16,452	Higher base credit
BPA phase-out	—	16,452 → 14,829	Credit shrinks above $181k
CPP YMPE	$71,300	$74,600	Higher CPP cap
CPP YAMPE	$81,200	$85,000	Higher CPP2 ceiling
Max CPP1 contrib	$4,034.10	$4,230.45	More CPP1 over the year
Max CPP2 contrib	$396.00	$416.00	Slightly more CPP2
EI MIE (max insurable)	$65,700	$68,900	Higher EI cap
EI employee rate	1.64 %	1.63 %	Slightly lower EI
Max EI premium	$1,077.48	$1,123.07	Slightly more EI YTD
Alberta brackets	5 brackets, 10–15 %	6 brackets, 8–15 %	NEW 8 % on first $61,200
BC lowest rate	5.06 %	5.60 %	Higher BC tax (Feb 17 budget)
Manitoba indexation	1.7 %	0 % (paused)	Brackets unchanged
NL Basic Personal Amount	$11,067	$15,000	Big bump
SK Basic Personal Amount	$18,491	$20,381	Affordability Act

All values verified Dec 2025 – Jan 2026 against CRA T4127, CPB Canada CPP announcement, the EI Premium Notice, and TaxTips.ca province-by-province tables.

1. Pay frequency math

Every employee has a pay frequency. The engine divides annual salary by periods/year to get the per-period gross before any deductions.

Frequency	Periods / year	Symbol	Example — $80,000 / yr
Weekly	52	weekly	$1,538.46
Bi-weekly	26	biweekly	$3,076.92
Semi-monthly	24	semimonthly	$3,333.33
Monthly	12	monthly	$6,666.67

Per-period gross is the starting point for CPP, EI, federal tax, and provincial tax. Each of those four services has its own way of annualizing it (CPP and EI subtract a per-period exemption first; federal/provincial multiply gross by periods to find the bracket).

2. CPP — Canada Pension Plan (2026)

Constants (CRA confirmed)

YBE — Year's Basic Exemption	$3,500.00
YMPE — Year's Maximum Pensionable Earnings	$74,600.00
YAMPE — Year's Additional Maximum Pensionable Earnings	$85,000.00
CPP1 employee/employer rate	5.95 %
CPP2 rate (between YMPE and YAMPE)	4.00 %
Max CPP1 contribution (annual)	$4,230.45
Max CPP2 contribution (annual)	$416.00

Formula (per pay period)

```
CPP1 = max(0, gross − YBE / periods_per_year) × 5.95 %
capped so YTD ≤ $4,230.45
```

```
CPP2 = (overlap of [YTD_pensionable, YTD+gross]
with [YMPE, YAMPE]) × 4.00 %
capped so YTD ≤ $416.00
```

Worked example — Ontario, $92,000 salary, bi-weekly

Step	Calculation	Value
Per-period gross	92,000 ÷ 26	$3,538.46
Per-period YBE	3,500 ÷ 26	$134.62

Taxable for CPP	$3,538.46 – $134.62	$3,403.85
CPP1 contribution	× 5.95 %	$202.53
CPP2 contribution	$92k < YMPE → no CPP2	$0.00

3. EI — Employment Insurance (2026)

Constants

MIE — Max Insurable Earnings	$68,900.00
Employee rate	1.63 %
Employer multiplier	1.4×
Max employee premium (annual)	$1,123.07
Max employer premium (annual)	$1,572.30

Formula

```
EI_employee = gross × 1.63 % (capped so YTD ≤ $1,123.07)
EI_employer = EI_employee × 1.4
```

Worked example — $92,000 salary, bi-weekly

Step	Calculation	Value
Per-period gross	92,000 ÷ 26	$3,538.46
EI employee premium	× 1.63 %	$57.68
EI employer cost	× 1.4	$80.75
Annual employee max	MIE × rate	$1,123.07

4. Federal income tax (2026)

2026 NEW: The lowest bracket dropped from 15 % to **14 %**. This lower rate also applies to the BPA credit and to the K2 (CPP+EI) credit.

Tax brackets

Taxable income (annual)	Rate	Tax at top of bracket
First $58,523	14.0 %	$8,193.22
$58,523 – $117,045	20.5 %	$20,190.23
$117,045 – $181,440	26.0 %	$36,932.93
$181,440 – $258,482	29.0 %	$59,275.11
Above $258,482	33.0 %	—

Basic Personal Amount — with phase-out

The BPA is a tax credit applied at the lowest bracket rate (14 %).

Annual income	Effective BPA
≤ $181,440	$16,452.00
$181,440 → $258,482 (linear taper)	$16,452.00 → $14,829.00
≥ $258,482	$14,829.00

```
effectiveBPA(income) = bpaHigh − (bpaHigh − bpaLow) × t
where t = (income − 181,440) / (258,482 − 181,440)
```

Per-period federal tax formula

```
annual_taxable = period_gross × periods_per_year
gross_annual_tax = sum of tax in each bracket
bpa_credit = effectiveBPA(annual_taxable) × 14 %
k2_credit = (period CPP + CPP2 + EI) × 14 % ← per period
period_fed_tax = max(0, (gross_annual_tax − bpa_credit) / periods − k2_credit)
```

5. Provincial income tax — all 9 provinces (2026)

Each province has its own brackets, BPA, and (Ontario only) surtax. The engine reads from a single map keyed by province code; the math is identical to federal except (1) no BPA phase-out and (2) Ontario adds a surtax on top of net tax.

ON — Ontario

Up to	Rate
$53,891.00	5.05 %
$107,785.00	9.15 %
$150,000.00	11.16 %
$220,000.00	12.16 %
no cap	13.16 %

Basic Personal Amount: $12,989.00 · **Overtime threshold:** 44 hrs / week · **Surtax:** 20 % over $5,818, +36 % over $7,446 (tax-on-tax)

AB — Alberta

Up to	Rate
$61,200.00	8.00 %
$154,259.00	10.00 %
$185,111.00	12.00 %
$246,813.00	13.00 %
$370,220.00	14.00 %
no cap	15.00 %

Basic Personal Amount: $22,769.00 · **Overtime threshold:** 44 hrs / week

BC — British Columbia

Up to	Rate
$50,363.00	5.60 %
$100,728.00	7.70 %
$115,648.00	10.50 %
$140,430.00	12.29 %
$190,405.00	14.70 %
$265,545.00	16.80 %

no cap	20.50 %

Basic Personal Amount: $13,216.00 · **Overtime threshold:** 40 hrs / week

MB — Manitoba

Up to	Rate
$47,564.00	10.80 %
$101,200.00	12.75 %
no cap	17.40 %

Basic Personal Amount: $15,780.00 · **Overtime threshold:** 40 hrs / week

SK — Saskatchewan

Up to	Rate
$54,532.00	10.50 %
$155,805.00	12.50 %
no cap	14.50 %

Basic Personal Amount: $20,381.00 · **Overtime threshold:** 40 hrs / week

NS — Nova Scotia

Up to	Rate
$30,995.00	8.79 %
$61,991.00	14.95 %
$97,418.00	16.67 %
$157,124.00	17.50 %
no cap	21.00 %

Basic Personal Amount: $11,932.00 · **Overtime threshold:** 48 hrs / week

NB — New Brunswick

Up to	Rate
$52,332.00	9.40 %
$104,666.00	14.00 %
$193,861.00	16.00 %
no cap	19.50 %

Basic Personal Amount: $13,664.00 · **Overtime threshold:** 44 hrs / week

PE — Prince Edward Island

Up to	Rate
$32,656.00	9.80 %
$81,310.00	13.80 %
no cap	16.70 %

Basic Personal Amount: $14,525.00 · **Overtime threshold:** 48 hrs / week

NL — Newfoundland & Labrador

Up to	Rate
$44,678.00	8.70 %
$89,355.00	14.50 %
$159,529.00	15.80 %
$223,340.00	17.80 %
$285,318.00	19.80 %
$570,635.00	20.80 %
$1,141,272.00	21.30 %
no cap	21.80 %

Basic Personal Amount: $15,000.00 · **Overtime threshold:** 40 hrs / week

6. Vacation pay — accrue vs payout

Every employee has a vacation rate (default 4 %, the federal minimum). **How** vacation is paid depends on the per-employee *vacationMode* field.

Mode: payout (default)

`vacation_amount = (regular + overtime + bonus) × vacation_pct`

`gross = regular + overtime + bonus + vacation_amount`

Vacation lands in **gross pay**, gets taxed this period. Used for most hourly employees.

Mode: accrue (banked)

`vacation_amount = (regular + overtime + bonus) × vacation_pct`

`gross = regular + overtime + bonus ← no vacation in gross`

`vacation_banked_this_period = vacation_amount`

Vacation accumulates in a separate bank, untaxed for now. Paid out later as a lump sum (taxed at that time).

Worked example — Ontario hourly, $30/hr, 80 hrs bi-weekly, 4 %

	Payout mode	Accrue mode
Regular pay	$2,400.00	$2,400.00
Vacation amount (4 %)	$96.00	$96.00
Gross (this period)	$2,496.00	$2,400.00
Vacation banked	—	$96.00
Taxed in this period?	Yes	No — taxed when paid out

7. Overtime

OT pays at **1.5×** the hourly rate. The trigger is weekly hours over the provincial threshold (column 'OT after').

Province	OT after (hrs/week)	Rate
ON	44	1.5×
AB	44	1.5×
BC	40	1.5×
MB	40	1.5×
SK	40	1.5×
NS	48	1.5×
NB	44	1.5×
PE	48	1.5×
NL	40	1.5×

Salaried OT — hourly equivalent

Salaried employees logged with OT hours get paid at an hourly-equivalent rate:

```
hourly_equiv = annual_salary / (standard_weekly_hours × 52)
ot_pay = hourly_equiv × 1.5 × ot_hours
```

Default standard week is 40 hrs (so annual / 2,080). Configurable per employee.

Example

Salary	Std weekly hrs	Hourly equivalent	5 OT hrs @ 1.5×
$80,000.00	40	$38.46	$288.46
$80,000.00	37.5	$41.03	$307.69
$120,000.00	40	$57.69	$432.69

8. Year-to-date enforcement

Each payroll run computes YTD per employee by folding all previously **finalized** runs in the same tax year. YTD then flows into CPP1, CPP2, and EI so caps stop deductions automatically mid-year.

What gets capped

Service	Annual cap	Behaviour after cap
CPP1	$4,230.45	Stops contributing
CPP2	$416.00	Stops once YTD pensionable ≥ YAMPE
EI employee	$1,123.07	Stops contributing
Federal tax	no cap	Continues per bracket
Provincial	no cap	Continues per bracket

Voided runs

When a payroll is voided, the original is kept (status='voided') and a reversal run is appended with negated line amounts. YTD always nets them to zero — no special case in the service.

9. Full worked example — Ontario, $92,000 salary, bi-weekly

End-to-end calculation. This is exactly what the engine produces.

Step	Calculation	Amount
1. Per-period gross	92,000 ÷ 26	$3,680.00
2. CPP1	(gross – 3500/26) × 5.95 %	$210.95
3. CPP2	92k < YMPE → 0	$0.00
4. EI	gross × 1.63 % (capped at YTD $1,123.07)	$59.98
5. K2 credit	(CPP + EI) × 14%	$37.93
6. Federal tax	see formula §4, less K2	$481.57
7. Ontario tax	see formula §5, includes surtax	$227.06
8. Total deductions	CPP + CPP2 + EI + Fed + ON	$979.56
9. Net pay	**gross – deductions**	**$2,700.44**

Employer cost this period: gross + CPP match ($210.95) + EI 1.4× ($83.97) = **$3,974.92**

10. Same employee at every pay frequency

Ontario, $80,000 salary. Different frequencies produce different per-period values but the same annualized totals — modulo rounding.

Frequency	Periods	Gross	CPP	EI	Fed	ON	Net
weekly	52	$1,600.00	$91.20	$26.08	$194.13	$91.29	$1,197.30
biweekly	26	$3,200.00	$182.39	$52.16	$388.26	$182.59	$2,394.60
semi-monthly	24	$3,466.66	$197.59	$56.51	$420.63	$197.80	$2,594.13
monthly	12	$6,933.34	$395.18	$113.01	$841.25	$395.61	$5,188.29

Annual gross check: weekly × 52 = $80,000.00 · biweekly × 26 = $80,000.00 · semi-monthly × 24 = $80,000.00 · monthly × 12 = $80,000.00. All match.

11. Same salary across all 9 provinces

$80,000 salary, bi-weekly. Federal/CPP/EI are identical; only the provincial tax line differs — and the resulting net pay.

Province	Gross	Federal	Provincial	CPP	EI	Net pay
ON	$3,200.00	$388.26	$182.59	$182.39	$52.16	**$2,394.60**
AB	$3,200.00	$388.26	$202.86	$182.39	$52.16	**$2,374.33**
BC	$3,200.00	$388.26	$177.26	$182.39	$52.16	**$2,399.93**
MB	$3,200.00	$388.26	$306.78	$182.39	$52.16	**$2,270.41**
SK	$3,200.00	$388.26	$275.74	$182.39	$52.16	**$2,301.45**
NS	$3,200.00	$388.26	$378.66	$182.39	$52.16	**$2,198.53**
NB	$3,200.00	$388.26	$306.01	$182.39	$52.16	**$2,271.18**
PE	$3,200.00	$388.26	$338.72	$182.39	$52.16	**$2,238.47**
NL	$3,200.00	$388.26	$314.14	$182.39	$52.16	**$2,263.05**

Highest net here: **Alberta** (new 8 % bottom bracket on first $61k + big BPA). Lowest: **Nova Scotia** (BPA only $11,932 + 14.95 % bracket starts at $30,995).

12. Hourly example with OT and vacation payout

BC, $30/hr, 88 regular hours + 4 OT hours bi-weekly, 4 % vacation (payout). Provincial OT threshold for BC is 40 hrs/week.

Step	Calculation	Amount
Regular pay	$30 × 88 hrs	$2,640.00
Overtime pay	$30 × 1.5 × 4 hrs	$180.00
Vacation (4 %)	(regular + OT) × 4 %	$112.80
Gross	sum of above	$2,932.80
CPP1	(gross – YBE/26) × 5.95 %	$166.49
EI	gross × 1.63 %	$47.80
Federal tax	see §4	$336.33
BC tax	see §5	$156.68
Net pay	**gross – all deductions**	**$2,225.50**

13. T4 box mapping

Every box on the T4 slip is computed from YTD totals over all finalized payroll runs in the tax year. Boxes not in this table are not yet populated.

Box	Label	Source
10	Province of employment	employee.province
12	Social Insurance Number	employee.sin
14	Employment income	Σ gross
16	Employee's CPP contributions	Σ CPP1
16A	Employee's second CPP contributions	Σ CPP2
17	Employee's QPP contributions	0 (no Quebec support)
17A	Employee's second QPP	0
18	Employee's EI premiums	Σ EI
22	Income tax deducted	Σ federal + provincial
24	EI insurable earnings	min(Σ gross, $68,900.00)
26	CPP pensionable earnings	min(Σ gross, $74,600.00)
28	Exempt (CPP/EI/PPIP) checkboxes	All unchecked
54	Employer's account number	company.businessNumber
Other	Codes 30, 40, 42, 66+ etc.	Empty (Phase 2)

14. Paystub PDF layout

Letter size, generated client-side via jsPDF, downloaded directly to the user's machine.

Section	Contains
Header band	Employer name, BN, CRA payroll account · Pay period · Pay date
Employee block	Name, SIN (masked), province, frequency, employment type, comp rate
EARNINGS table	Regular / OT (1.5×) / Bonus / Vacation paid — current + YTD columns
DEDUCTIONS table	Federal / Provincial / CPP / EI — current + YTD
NET DEPOSIT box	Large $ amount + YTD net + 'PAID' tag
Vacation banked	Only shown when employee.vacationMode = 'accrue'
Employer contributions	CPP match + EI 1.4× + total employer cost
Footer	CRA disclaimer + 'Generated by NorthPay'

15. Payroll lifecycle + validation

Every run flows through a state machine before being persisted:

`draft → preview → finalized → voided`

Only **finalized** runs contribute to YTD, paystubs, and T4s. Voided runs stay in the record and are netted to zero by their reversal entry.

Pre-finalization validation checks

Code	Severity	Triggers if…
PERIOD_INVALID_DATES	error	Non-parseable date strings
PERIOD_END_BEFORE_START	error	end < start
PAY_DATE_BEFORE_PERIOD_END	error	pay date earlier than period end
DUPLICATE_PAYROLL	error	Same employee + same period already finalized
EMPLOYEE_NAME_MISSING	error	Empty first or last name
EMPLOYEE_PROVINCE_UNSUPPORTED	error	Quebec or unknown province
EMPLOYEE_SALARY_INVALID	error	Salary type with ≤ 0 annual
EMPLOYEE_HOURLY_INVALID	error	Hourly type with ≤ 0 rate
EMPLOYEE_VACATION_UNUSUAL	warning	Vacation % outside 0–20 %
CPP_CAP_REACHED	warning	YTD CPP1 already at annual max
CPP2_CAP_REACHED	warning	YTD CPP2 already at annual max
EI_CAP_REACHED	warning	YTD EI already at annual max
CPP2_TIER_EXHAUSTED	warning	YTD pensionable > YAMPE

Errors block finalization. **Warnings** are surfaced in the UI but the run still completes.

Out of scope (intentionally not modeled yet)

Item	Reason
Quebec (QPP/QPIP/RAMQ)	Separate provincial system
Statutory holiday averaging	Provincial floor; needs holiday calendar
TD1 personal credits (spouse, disability, age)	Per-employee; not captured yet
RPP / DPSP contributions	Benefits — Phase 2
Taxable benefits (Box 40)	Benefits — Phase 2
Union dues (Box 44)	Benefits — Phase 2

ROE generation	Termination flow
T4 XML for e-filing	CRA T619 envelope — Phase 2
WSIB / WCB	Provincial workers' comp